

CPT208 Human-Centric Computing

06. Prototyping Fidelity and Dimensions

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Today's plan

1. Dimensions in prototyping
2. Physical design
3. Tips for prototyping

Fidelity in Prototyping (cont.)

Chapter 12

Questions first

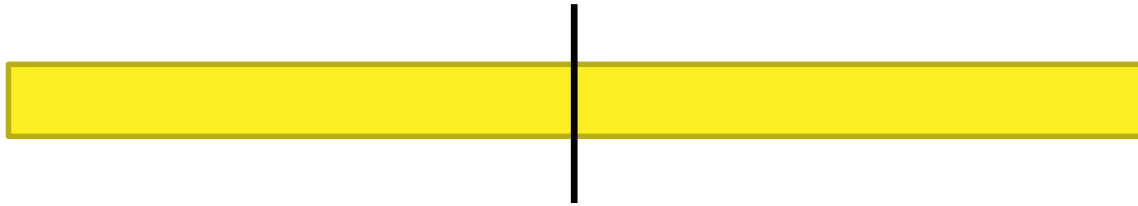
Are we using the low- or high-fidelity prototype if we used the colored UI/operable interface/coding...?

There is **no clear separations!**

But we do have measures from different perspectives.

fidelity is a **spectrum**

Is it something like this?



Or something like this?



How to define the level of fidelity

Or even this



Visual



Interaction



Breadth



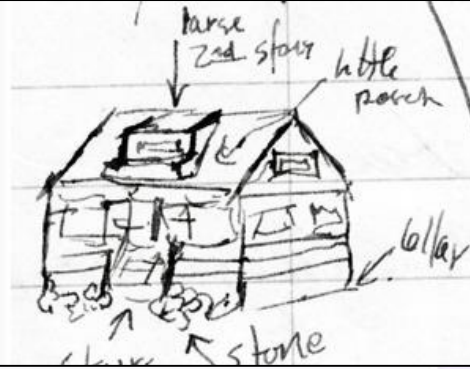
Depth



Content

Remember the building prototype?

Visual	
Interaction	
Breadth	
Depth	
Content	



Visual	
Interaction	
Breadth	
Depth	
Content	



Remember the **compromise!**

- **Neither** of these is inherently **good or bad**.
- There are **costs and trade-offs** with raising or lowering fidelity in any dimension.
- The path you take as the designer to get from the first image to the last is **up to you to determine**.

Why it matters?

- We're **wasting time and sanity** with imprecise language about fidelity.
- Even worse, we risk falsely validating one thing today and failing to validate a better way tomorrow.
- Teams are prone to over-engineering a prototype prematurely, leading to wasted effort. In other words, a prototype may be low fidelity except for one crucial dimension that may require quality and fidelity at near-production levels to validate fairly.

Dimensions

Chapter 12

Five dimensions in prototyping

- **Visual:** How real does it look?
- **Interaction:** How real does it feel?
- **Breadth:** To what degree is this the whole or just a part?
- **Depth:** At a given level of breadth, to what degree is the user constrained?
- **Content:** How real is the stuff and is it contextual to the user?

Dimensions - Visual fidelity

**“I’ll know it when I see it.
I can see what you’ve done
there and I like it.”**



- We judge books by their **covers**, even when we’re told not to — because we can’t help it.
- It is the **most direct dimension** that user feels about the prototype

Dimensions - Visual fidelity

Refers to the devotion and granularity put into the design for **visual interfaces and physical objects**

- In early stages, the visual fidelity are rather low, so users will focus on the **actions/procedures/concepts..**
- In later stages, it should be higher to test the **accessibility/touch/visual details...**



Dimensions - Visual fidelity

- The **big, obvious problem** involves **jumping to high visual fidelity too early** in the process.
 - There's a feeling that when something looks done, it must be done.
- Just because a designer **is capable of prototyping fairly quickly** at high visual fidelity **does not make it the right decision**.
 - Designers should sometimes intentionally keep visual fidelity low to encourage the kind of communication and feedback they want.

Dimensions - Interaction fidelity

“I’ll know how what is the feedback if I click/tap/swap.”



→ **InVision** lets anyone stitch screens together to increase interaction fidelity. **Balsamiq** has screen-linking capabilities that combine with a low-visual-fidelity component library.

Dimensions - Interaction fidelity

Refers to the objective degree of **exactness with which real-world interactions can be reproduced**

- Users can **click, tap, and swipe** through screens and **get the feel**.
- It allowed the team to **quickly and cheaply test hundreds of variations**.

Dimensions - Interaction fidelity

→ The palm pilot

- The most creative use of a chopstick
- Imagine the era that there is no smart phone
- How would design and test **the way how people use portable computers?**



Dimensions - Breadth fidelity



Let's say we wanted to redesign Amazon.com. Where would we start?

- Should we just start by prototyping the home page and checkout pages?
- What about product pages, search results, filtering, or wish list pages?
- **Amazon.com is huge.** Choosing where to draw the lines of your prototype's edges sets up its **breadth**.

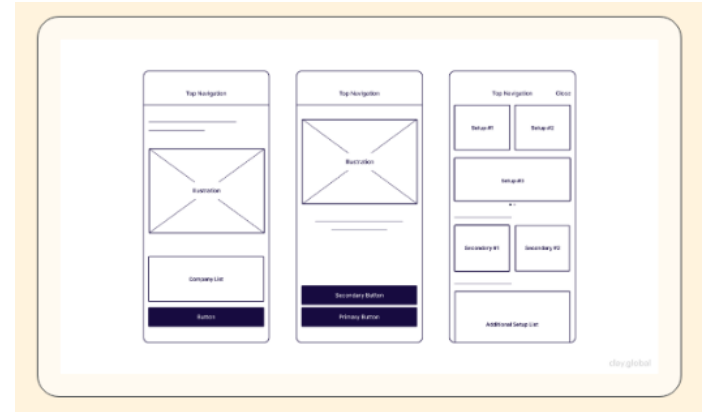
Dimensions - Breadth fidelity

Simply speaking, it refers to **the scope of features** available.

- Sometimes you will want to prototype it all.
Sometimes you should avoid most of the product's surface area because it can't change anyway.
- It's up to you to decide because **the entry and exit points** to your redesign may matter more than you think.

Horizontal Prototyping

- A **horizontal prototype** models the surface layer of the entire user interface.
- Users can navigate through top-level menus and view the layout of major screens, but attempting to perform a specific action will usually result in a static confirmation screen or a dead end.
- **Primary Focus:** Evaluating the Information Architecture (IA) and overall navigation paradigm before committing to complex backend development.



Strategic Uses for Horizontal Prototyping

→ Testing Navigation & Categorization

- If your app has dozens of features, horizontal prototypes prove whether users intuitively know where to look. E.g., Testing if users instinctively search for "Privacy" under "Settings" or "Profile."

→ Stakeholder Alignment & Buy-In

- It provides clients and management with a tangible "feel" for the entire software's scope very early in the conceptual phase, managing expectations before coding begins.

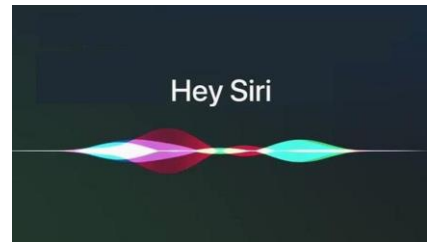
→ Rapid Conceptual Iteration

- Because functional depth is absent, designers can completely overhaul the navigation structure overnight based on user feedback without breaking backend databases.

Dimensions - Depth fidelity

Imagine that this time, we're prototyping a voice assistant (e.g., Siri, Alexa).

- Pretend we **limited the breadth** of our voice assistant prototype to a single domain: **weather**.
- If I can ask about **anything at all related to weather** and get a response, that's high depth fidelity — in other words, it's **deep**.
- If I can only inquire about forecasts in major cities, that's less depth. If I can ask only about my current location's temperature, that's even lower.

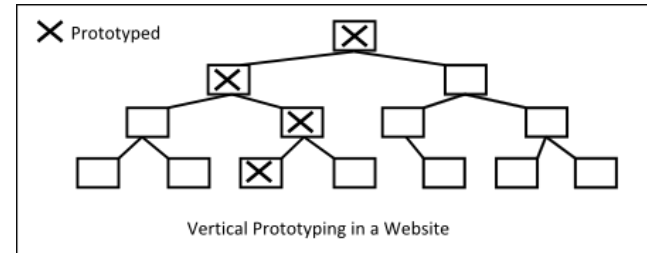


Dimensions - Depth fidelity

- Simply speaking, it refers to the **degree of functionality** within a single feature.
- Remind yourself about depth by asking **if the interactions are shallow.**
- How **significant** are my constraint handcuffs?

Vertical Prototyping

- A **horizontal prototype** models the surface layer of the entire user interface.
- It acts as a vertical slice through the application's layers—from the front-end user interface down to the backend logic, and occasionally the database.
- While the main navigation might be missing or unclickable, the user can successfully complete a highly complex task (e.g., a multi-step checkout process with real validation errors).



Strategic Uses for Vertical Prototyping

→ Evaluating Complex Task Usability

- Essential for testing high-risk flows where user friction leads to drop-offs (e.g., user registration, complex data entry forms, or financial transactions).

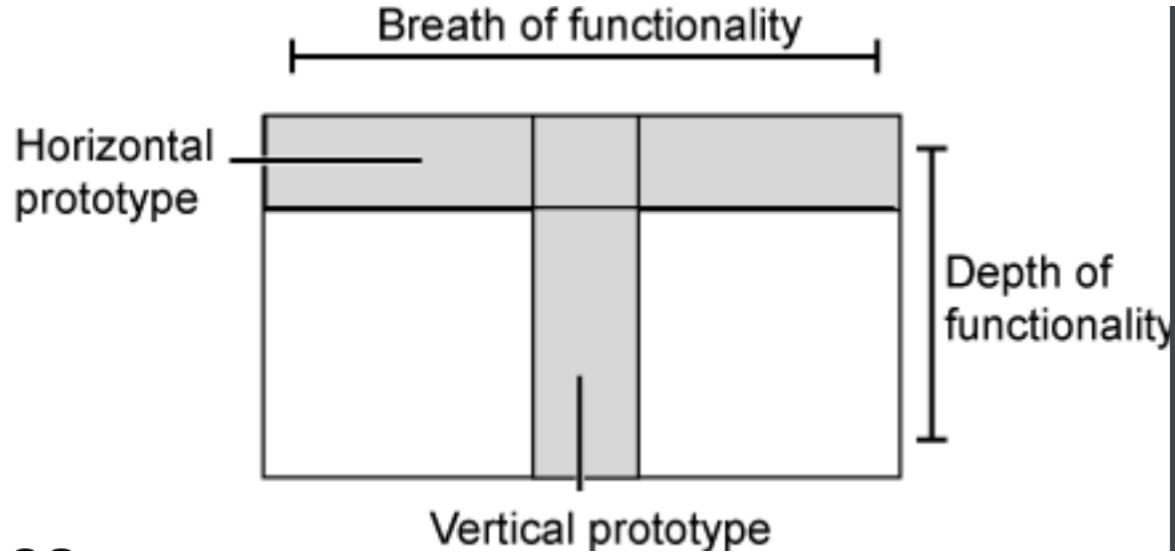
→ Assessing Technical Feasibility

- It allows developers to prove that a complex algorithm or database query actually works in real-time before applying that architecture to the rest of the application.

→ Testing Error Recovery States

- Because the logic is deep, researchers can intentionally trigger system errors (e.g., "invalid credit card") to see if the interface successfully guides the user to a resolution.

Dimensions - Breadth and Depth



Dimensions - Content fidelity

“No one reads the user notice.”



Lorem
ipsum

- Content is the **king**. The video, the data, and the names of their teams are what customers care about in our tools.
- Make good decisions in showing **the right content** in prototypes

Dimensions - Content fidelity

Refers to the **degree of realism** for the content.

- Imagine your prototype about serious recruitment apps with fake names as SpongeBob
- The content fidelity will impact on the feedback from users



Dimensions - Content fidelity

→ Have you ever confused someone with **lorem ipsum** when you hoped they would comment on your impeccable use of negative space?

Lorem Ipsum

"Neque porro quisquam est qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit..."

"There is no one who loves pain itself, who seeks after it and wants to have it, simply because it is pain..."

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Donec at tincidunt sapien. Maecenas semper dapibus erat, non porta sem volutpat a. Etiam vehicula viverra libero, vel eleifend nulla commodo sit amet. Proin imperdiet nibh dapibus dolor tincidunt, et euismod justo consectetur. Proin cursus tellus metus, elementum porttitor justo tempus ac. Nullam porttitor ullamcorper molestie. Mauris volutpat euismod dolor nec aliquam. Quisque suscipit erat quis risus tempus, sed posuere ipsum sagittis. In tortor lacus, tristique ac convallis rhoncus, sodales eu mi. Quisque mollis elementum metus vitae rutrum. Nam rutrum nulla ac lacus suscipit tempus. Pellentesque et tortor id odio tincidunt congue. Phasellus convallis libero dolor, id efficitur nisi dapibus quis. Nulla ornare arcu in orci suscipit, at volutpat purus mollis.



lorem
ipsum

OTHER dimensions

→ Autonomy:

→ operates alone vs requires
“supervision”

→ Platform:

→ interim vs final implementation

IMPORTANT LESSONS:

- it is **COMPLICATED** (slow, expensive) to prototype multiple dimensions at once.
 - so don't. Instead: modularity of prototyping.
- Each prototyping tool has **strengths and weaknesses**
 - may be better (more efficient and capable) for some of these prototyping dimensions than others.
 - you may need multiple tools throughout your design's life cycle.

Concrete Design

Chapter 12

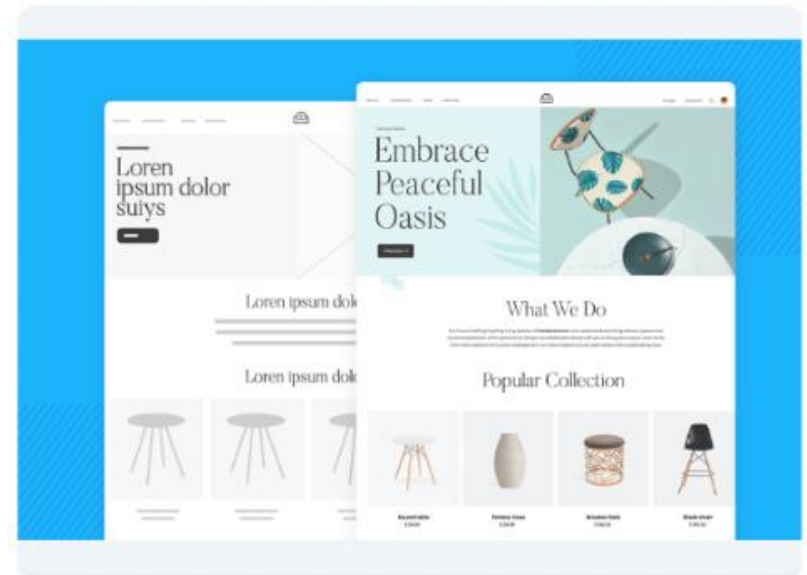
Getting concrete (vs. conceptual design)

- Considers **more concrete, detailed issues** of designing the interface
- **Iteration** between concrete and conceptual design
- From "**What the system does**" to "**How the user interacts**"

Getting concrete

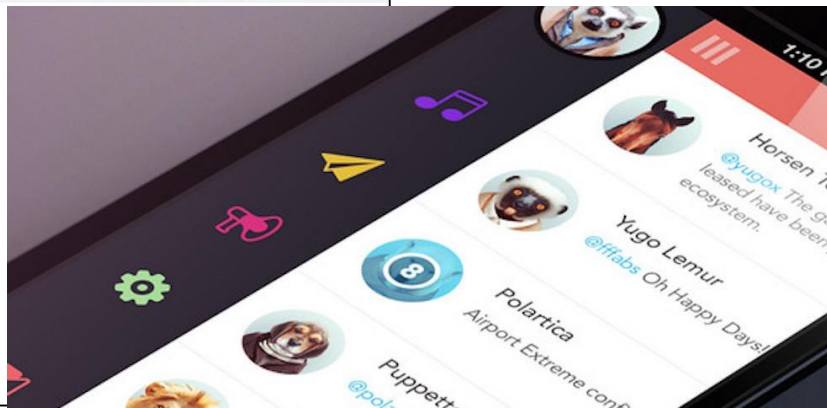
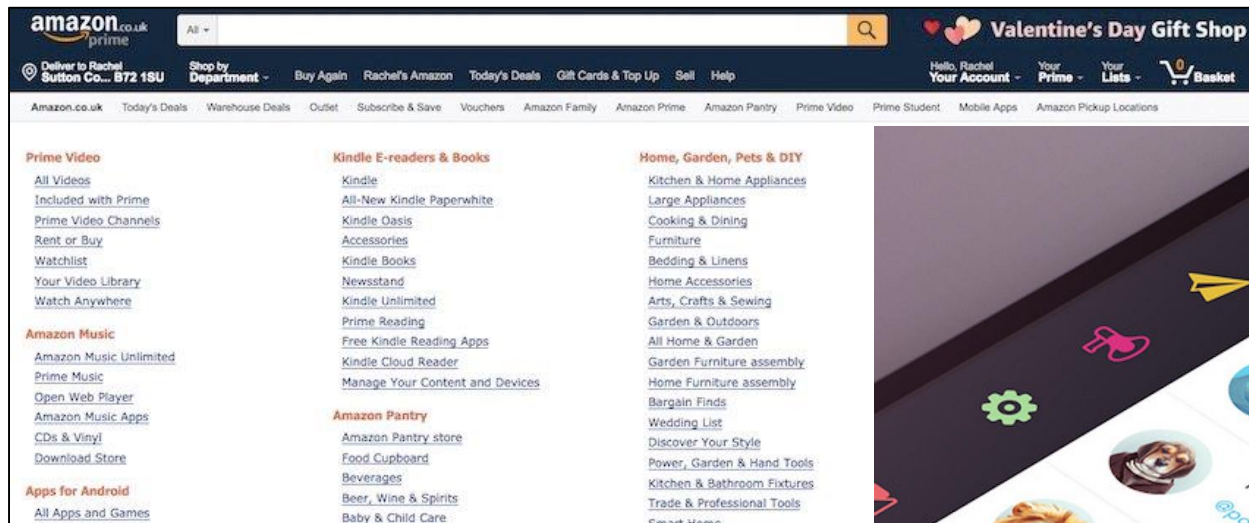
→ Different kinds of widget (dialog boxes, toolbars, icons, menus etc.)

- menu design
- icon design
- screen design
- information display



Menu design

- Structuring **navigation choices into specific formats** (dropdowns, accordions, mega-menus) based on context.



Menu design

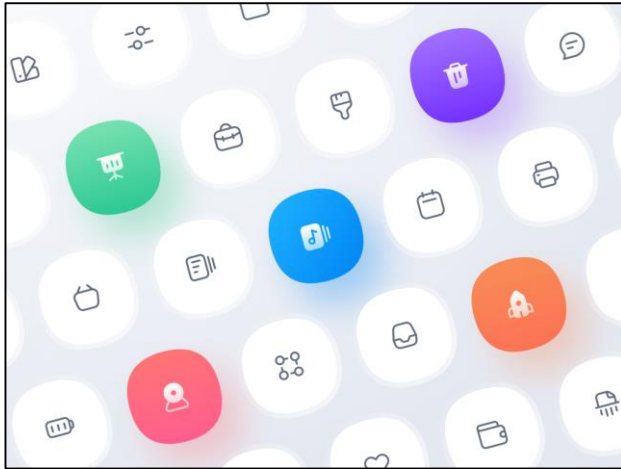
- **How long/deep** is the menu?
- **In what order** will the items appear?
- How is the menu to be **structured**, e.g., when to use sub-menus, dialog boxes?
- **What categories** will be used to group menu items?

Menu design

- How will **division into groups** be denoted, e.g., different colors, dividing lines?
- **How many** menus will there be?
- **What terminology** to use?
- How will any **physical constraints** be accommodated, e.g., mobile phone?

Icon design

- Creating **visual metaphors that communicate functionality instantly**, balancing style with universal recognition.

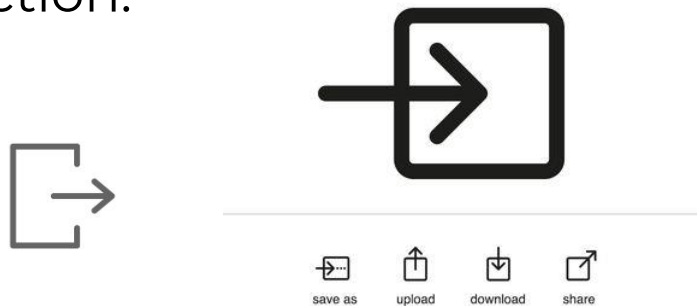


Icon design

- Good icon design is **difficult**
- Meaning of icons is **cultural and context sensitive**
- Some tips:
 - always draw on **existing traditions or standards**
 - **concrete objects or things** are easier to represent than actions

Icon design

- Avoid excessive detail. Icons must remain legible when **scaled down** to 16x16 pixels on mobile screens. Choose flat or simple vector designs.
- When in doubt, **label it**. A recognized icon paired with a concise text label dramatically increases usability and reduces cognitive friction.



Screen design

→ Two aspects:

→ **How to split across screens**

- moving around within and between screen
show much interaction per screen? serial or
workbench style?

→ **Individual screen design**

- white space: balance between enough
information/interaction and clarity
- grouping items together: separation with
boxes? lines? colors?

Splitting functions across screens

- **Task analysis** as a starting point
 - Each screen contains a single simple step?
 - **Frustration** if too many simple screens
- Keep **information available**
 - multiple screens open at once

Splitting functions across screens



Jisc: Assent Liberate

My profile Help Logout

Dashboard Service view Liberate configuration Organisation Camford University

Camford University Liberate configuration

General Policies SAML IdP Web proxy ABFAB IdP Eduroam

General configuration

WARNING Your general configuration is incomplete and xxxxxx outline of consequences xxxxxx

Directory server access

DNS domain name * Validation error message here

example.com

LDAP/AD service account *

Password *

Retype password * Passwords match

Directory server advanced settings

LDAP encryption ☐ None ☒ LDAP / StartTLS ☐ LDAP

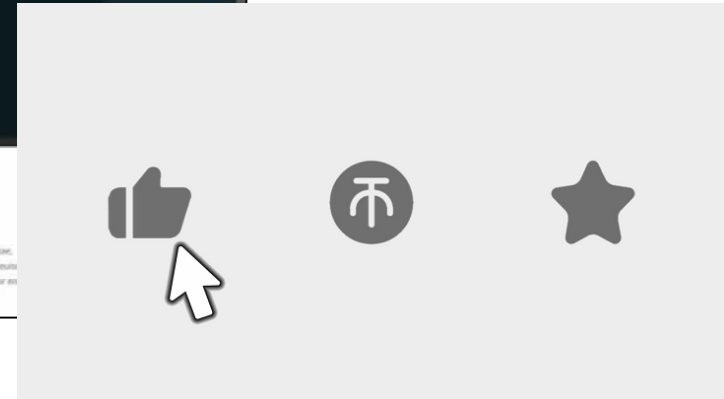
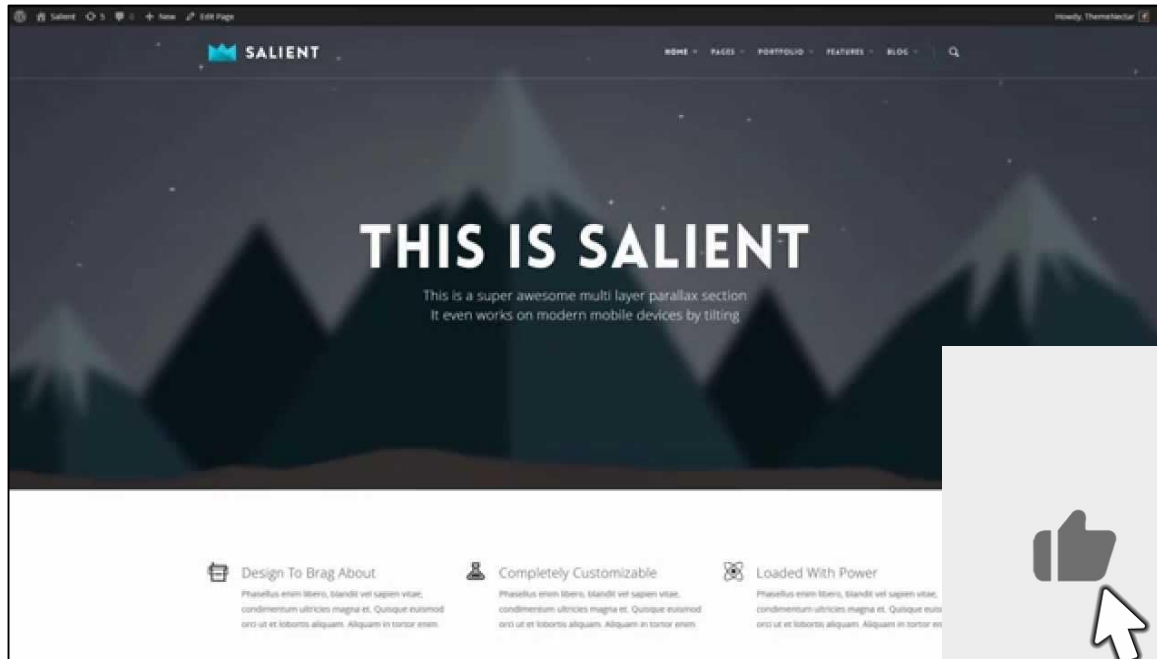
LDAP/AD server TLS CA * Browse Save the certificate

Use default LDAP port? ☐ Yes ☒ No

individual screen design

- Draw **user attention** to **salient point**, e.g., colour, motion, boxing (Visibility?)
 - **Animation** is very powerful but can be distracting
 - **Good organization** helps: grouping, physical proximity
- **Trade off** between sparse population and overcrowding

individual screen design



Information display

- **Relevant** information available at all times
- Different types of information imply **different kinds of display**
- **Consistency** between paper display and screen data entry

Information display



Any Questions?

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